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Submitted By

Ranga.Sai Manoj – 22091A05D7(RGMCET,4th year)

Goruparthi.Spoorthi-22HP1A1223(ALIET,4th year)

Vangala.Chaitanya Kumar Reddy-22091A0521(RGMCET,4th year)

Siddabathula.Tejaswi-18F01A0242(st Anns clg, passed out)

B.Lokesh-22NU1A0411(NSRIT,4th year)

Under the guidance of

Meenakshi

V. Vanitha

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ABSTRACT:

The Indian job market is evolving rapidly, shaped by digital transformation, economic shifts, and changing skill demands. In this project, we used Python to explore and analyze job-related data to better understand current employment trends across the country.

We used the help of libraries like Pandas, NumPy, Matplotlib, and Seaborn, we cleaned and visualized the data to uncover patterns in job roles, salaries, locations, and required skills.

Our findings highlight a growing demand for tech-driven roles such as software developers and data analysts form various cities.

This study not only offers insights for job seekers and recruiters but also demonstrates how Python can be a powerful tool for making sense of complex labor market data.

Primary Objectives for analysis of the India job Market:

Understand Employment Trends:

Track changes in employment rates and job creation over time Identify growing and declining sectors (e.g., IT, manufacturing, agriculture), analyze full-time vs. part-time or gig economy shifts.

Assess Skill Demand and Gaps:

Map current and future skill requirements across industries, pinpoint mismatches between education/training and industry needs, shape upskilling and reskilling policies accordingly.

Monitor Demographic and Regional Variations:

Examine employment patterns across age, gender, caste, and region, spotlight rural vs. urban employment challenges and opportunities, policy interventions to local needs.

Forecast Future Workforce Needs:

Anticipate labor market changes from tech adoption (AI, automation),project demand for green jobs, digital services, and startup growth.

Support Data-Driven Decision Making:

Equip policymakers, businesses, and educators with actionable insights, help job seekers make informed career decisions based on trends and demand.

Empower Businesses and Job Seekers

Help companies optimize recruitment strategies and workforce planning, enable job seekers to make informed career decisions based on real-time market data.

Throughout, the analysis we employ the python’s library to clean, preprocess and then manipulate the data ,ensuring the accurate results for the analysis, we also import the numpy, matplotlib.pyplot (box plot, scatterplot, bar-graphs, histography, pie-charts etc.) ,pandas and the sciklearn for understanding and analysing the complex data and patterns.

Analyzing India’s job market helps identify high-growth sectors, assess employment quality, and uncover skill mismatches. It reveals regional and demographic disparities, guiding inclusive workforce policies. The analysis also supports forecasting future job trends and evaluating the impact of government initiatives. These insights enable data-driven decisions that foster sustainable employment and economic resilience.

Introduction:

In the dynamic and competitive world, Analyzing the Indian job market using Python offers a powerful, data-driven approach to understanding employment trends, skill demands, and sectoral shifts across the country. With its rich ecosystem of libraries like pandas, NumPy, Matplotlib, and Seaborn.

By harnessing the power of Python's versatile libraries such as Pandas and Matplotlib, this study presents a comprehensive analysis of india_job_market data, aiming to unlock hidden patterns and trends that can drive strategic decision-making for development of skills and the required trends.

The dataset used in this analysis comprises a wealth of information, including job title, salary, education, skills required, job portal, onsite/offsite, and many more.

Through Python's Pandas library, we perform data preprocessing, cleansing, and transformation, ensuring data accuracy and consistency. Leveraging Pandas' rich functionality, we delve into exploring skills and demand gaps, driven decision making, future workforce needs etc.

Furthermore, we employ Matplotlib, a powerful data visualization library, to create intuitive and informative plots, charts, and graphs.

These visual representations aid in presenting complex data in a compelling and easy-to-understand manner, facilitating the communication of key findings to stakeholders and decision-makers.

By analysing job_market data using Python, Matplotlib, and Pandas, we aim to provide job requriments with data-driven insights to optimize their business strategies. Armed with these insights, job market can make informed decisions, adapt to changing market demands, and enhance their organizations overall experience.

Sectoral Hotspots:

Several sectors are driving significant job growth.

Information Technology (IT) and IT-Enabled Services (ITeS): Continues to dominate, with a strong focus on digital transformation, cloud, AI, and cybersecurity.

Healthcare and Pharmaceuticals:

Booming due to increased spending, advancements in biotechnology, and telemedicine.

Financial Services and FinTech:

Expanding rapidly with digital banking, mobile payments, and blockchain

E-commerce and Retail:

Driven by the growth of online shopping, creating demand in logistics, supply chain, digital marketing, and customer service.

Manufacturing and Engineering:

Benefiting from government initiatives like "Make in India," leading to increased demand for engineers and skilled labor.

Renewable Energy:

Poised for significant growth as India pursues ambitious sustainability goals.

Rise of the Gig Economy and Flexible Work:

The gig economy is expanding, with more individuals choosing freelance and contract work due to online platforms offering greater autonomy and work-life balance. Remote and hybrid work models, accelerated by the pandemic, are also becoming integral to the job market.

High Job Mobility:

Indian professionals are exhibiting high job mobility, actively seeking new roles and negotiating for better pay and benefits. They are increasingly prioritizing purpose, ethics, and flexibility in their careers

This overall study showcases the potential of data analysis and visualization tools in the hospitality industry, illustrating their practical applications in improving operational efficiency and maximizing revenue generation. As such, the findings presented here serve as a valuable resource for organizations, analysts, and researchers seeking to unlock the hidden potential within their data and elevate their performance in the competitive landscape.

SYSTEM REQUIREMENTS:

SOFTWARE REQUIREMENT:

- **OPERATING SYSTEM:**

  The analysis can be performed on Windows.

- **Python:**

  Python 3.x is required for running the analysis. Make sure you have the latest stable version of Python installed.

- **Libraries:**

Pandas:

Install the Pandas library using pip, a package manager for Python.

Syntax: pip install pandas

Matplotlib:

Install the Matplotlib library using pip

Syntax: pip install matplotlib

Seaborn:

Install the seaborn library using pip command

Syntax: pip install seaborn

HARDWARE REQUIREMNTS:

- **IDE** – Jupyter Notebook, Google Collaboratory
- **Storage Space** – free storage space enough for running on machine

ARCHITECTURE:

The architecture of the Indian job Market analysis using Python, Matplotlib, and Pandas involves several key steps that form a cohesive workflow. The process typically includes data acquisition, data preprocessing, exploratory data analysis (EDA), data visualization.

Let's explore the architecture in more detail:

Data Acquisition:

The analysis begins with obtaining the Indian Job Market dataset. Data can be collected from various sources, such as a Organizations internal database, publicly available datasets, or through web scraping from different platforms.

Once the data is acquired, it is typically stored in a structured format like CSV, Excel, or a database.

Data Preprocessing:

Data preprocessing is a crucial step to ensure data quality and consistency.

Using Python's Pandas library, the data is loaded into a DataFrame, allowing for easy data manipulation and analysis.

This step involves handling missing values, handling duplicates, converting data types, and addressing any data quality issues.

Exploratory Data Analysis (EDA):

EDA involves exploring the data to gain insights into its structure, distributions, and relationships between variables.

Pandas' functions are used to perform summary statistics, groupings, and aggregations to understand key trends and patterns.

Data visualization with Matplotlib helps to create plots, histograms, scatter plots, and other visualizations to visualize patterns and correlations effectively.

Data Visualization:

Matplotlib is a powerful library for creating various types of visualizations, enabling the presentation of complex data in an intuitive and informative manner.

Visualizations are used to communicate key findings, such as Skill requirements over time, Job roles, education preferences, and deadline and openings of the applications.

Insights and Decision Making:

The final step is to draw meaningful insights from the analysis and make data-driven decisions to optimize the requirements for the role, marketing strategies, and organization trends.

The analysis results can be presented in reports, dashboards, or interactive visualizations to aid stakeholders in understanding the key takeaways effectively.

The Indian Job Market analysis using Python, Matplotlib, and Pandas follows a well-structured architecture that seamlessly integrates data acquisition, preprocessing, exploratory data analysis, visualization.

USES OF DATA ANALYSIS LIBRARY:

Pandas, Matplotlib, and Seaborn play crucial roles in the Indian Job Market analysis using Python, enabling a comprehensive and data-driven approach to understand skill requirements, organization preferences, and technology trends. Here's a detailed short note on their uses in this analysis:

Pandas:

- Data Manipulation: Pandas provides powerful data manipulation capabilities, enabling easy loading, cleaning, and preprocessing of the India Job Market dataset. It allows filtering, grouping, and aggregating data to derive meaningful insights.

- Data Exploration: Pandas facilitates the exploration of booking patterns over time, customer segmentation based on demographics, and analysis of skill requirements, organization preferences, and revenue metrics.

- Handling Missing Data: Pandas' functions handle missing data points effectively, ensuring data quality and preventing biases in the analysis.

- Data Transformation: It aids in transforming data into a format suitable for analysis, such as converting data types and applying mathematical operations.

- Joining and Merging: Pandas is used to combine datasets when additional information, such as organization reviews or Job amenities, is available separately.

Matplotlib:

- Data Visualization: Matplotlib allows the creation of various visualizations like line plots, bar charts, and scatter plots to depict skills, job roles, and companies distribution.

- Time Series Analysis: With Matplotlib, time series plots can illustrate salary, no of vacanies over specific time intervals, highlighting requirement variations and trends.

- Geospatial Analysis: Matplotlib can generate geographic maps that visualize customer distribution, providing insights into high-demand regions and popular Job locations.

Seaborn:

- Enhanced Data Visualization: Seaborn is built on top of Matplotlib and offers more aesthetically pleasing and informative visualizations. It simplifies the creation of complex plots like heatmaps, pair plots, and violin plots.

- Statistical Insights: Seaborn provides built-in statistical functions that allow us to easily visualize relationships between variables, such as correlation matrices or regression plots for revenue analysis.

- Categorical Data Visualization: It excels at visualizing categorical data, such as type of Job or organizations needs, using bar plots or box plots, which aids in understanding preferences and trends.

This combined toolkit empowers organizations to gain valuable insights from their skill requiremnts, sector growth, and improve the experiences for a competitive edge in the IT and Non-IT sectors .

Indian Job Market Queries and Outputs:

Importing Libraries:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Read DataSet:

```
ijm= pd.read_csv('ijmdataset.csv')
ijm
```

Basic Functions:

```
ijm.info()
ijm.shape
ijm.head()
ijm.tail()
ijm.unique()
ijm['Job Title'].value_counts()
ijm.isnull()
```

Data Cleaning:

```
1    ijm.drop_duplicates(inplace=True)
     uni_rows=ijm.shape[0]
     uni_rows
```

Output:

20000

```
2    ijm = ijm.dropna(subset=['Job Title'])
```

Data Filtering:

```
fulltime_h = ijm[(ijm['Job Type'] == 'Full-time') & (ijm['Number of Applicants'] > 100)]
fulltime_h.head()
```

Output:

Shows the top 5 rows from the data set.

```
recent = ijm[ijm['Posted Date'] > '2025-01-01']
recent.head()
```

Output:

Executes the recently uploaded top 5 rows from the dataset.

```
ijm.groupby('Job Type')['Number of Applicants'].mean()
```

Output:

Number of applications	
Job type	
Full time	**258.400000**
Part time	**256.800033**
internship	**245.675338**
contract	**256.303645**

dtype: float64

```
ijm.groupby('Company Name').size().rename('Job Listing')
```

Number of Applicants.	
Salary Range	
12-20 LPA	254.488411
20+ LPA	258.448088
3-5 LPA	258.899399
5-8 LPA	256.369297
8-12 LPA	255.245363

dtype: float64

Aggregations:

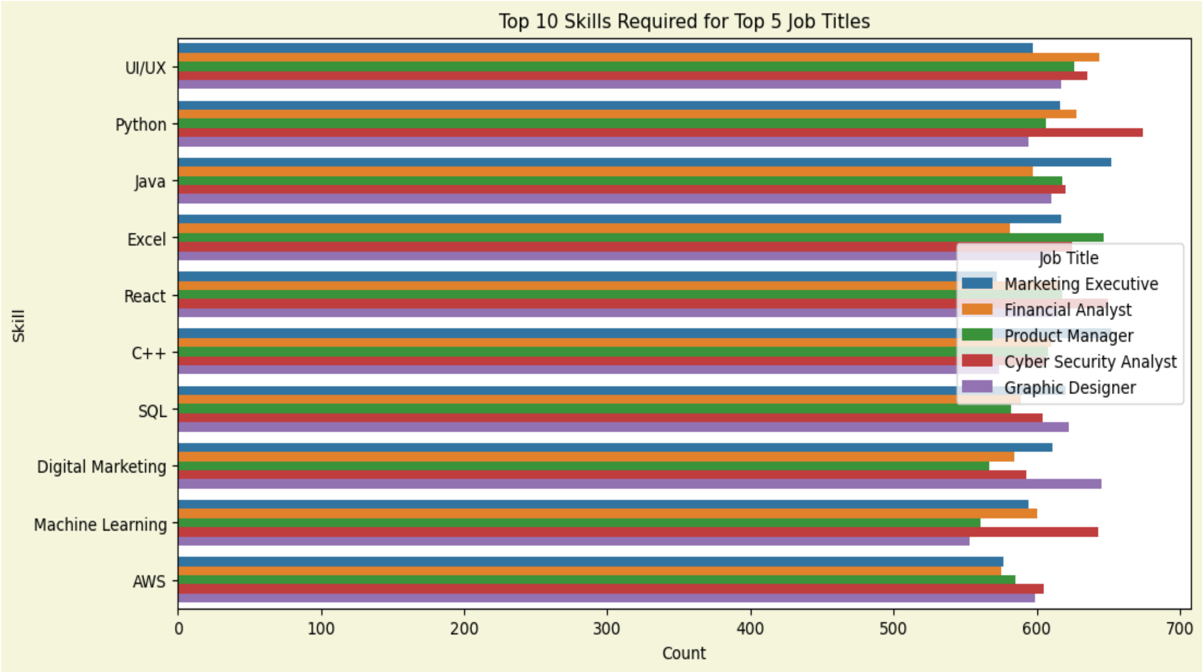
```
ijm['Number of Applicants'].min()

ijm['Number of Applicants'].max()

ijm['Number of Applicants'].mean()

print(int(ijm['Number of Applicants'].mean()))

ijm['Number of Applicants'].var()
```

Data Vizualization:

Horizontal Bar Graph:

```
top_jobs = ijm['Job Title'].value_counts().nlargest(5).index

filtered_df = ijm[ijm['Job Title'].isin(top_jobs)].copy()

filtered_df['Skills Required'] = filtered_df['Skills Required'].str.split(',')

filtered_df = filtered_df.explode('Skills Required')

filtered_df['Skills Required'] = filtered_df['Skills Required'].str.strip()

plt.figure(figsize=(12,6),facecolor='beige')

sns.countplot(y='Skills Required', hue='Job Title', data=filtered_df, order=filtered_df['Skills Required'].value_counts().index[:10])

plt.title('Top 10 Skills Required for Top 5 Job Titles')

plt.xlabel('Count')

plt.ylabel('Skill')

plt.legend(title='Job Title')

plt.show()
```

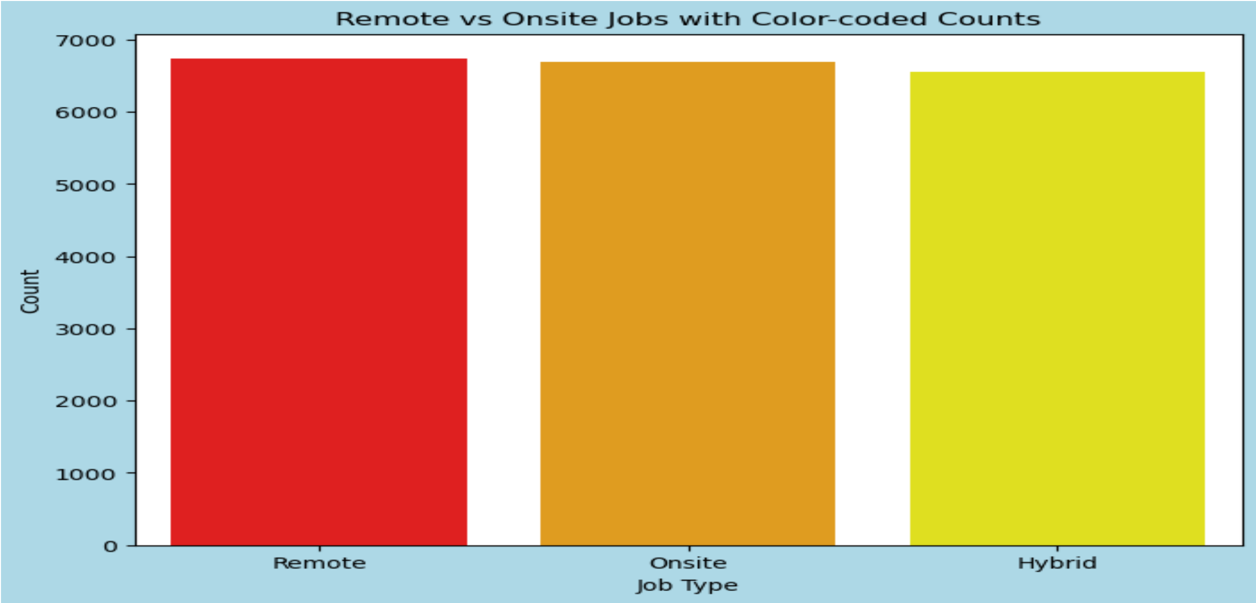
Output:



Vertical BarGraph:

```
counts = ijm['Remote/Onsite'].value_counts()

categories = counts.index

values = counts.values

colors = np.full(len(values), 'yellow', dtype=object)

colors[np.argmax(values)] = 'red'

if len(values) > 2:

    middle_pos = np.argsort(values)[-2]

    colors[middle_pos] = 'orange'

plt.figure(figsize=(8,6),facecolor='lightblue')

sns.barplot(x=categories, y=values, hue=categories, legend=False, palette=colors.tolist())

plt.bar(categories, values, color=colors.tolist())

plt.title('Remote vs Onsite Jobs with Color-coded Counts')

plt.xlabel('Job Type')

plt.ylabel('Count')

plt.show()
```

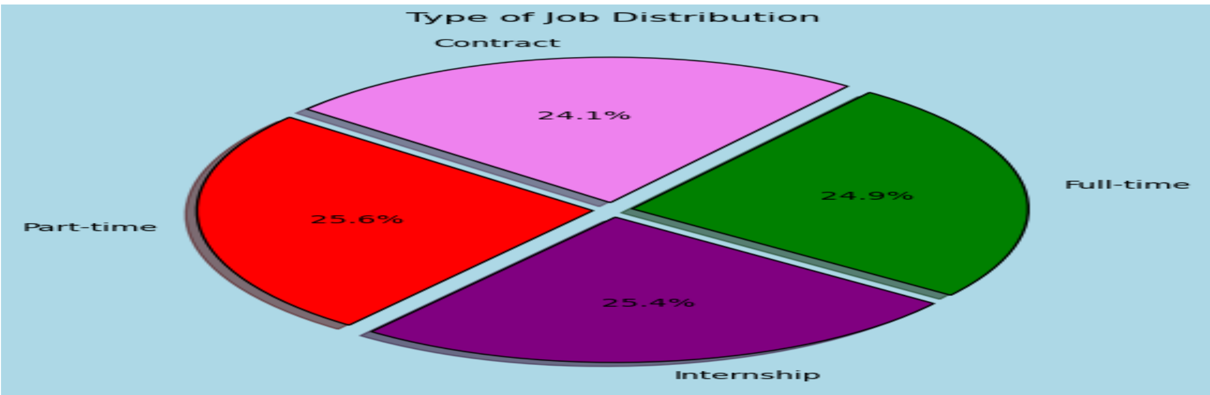
Output:



Pie-Chart:

```
plt.figure(figsize=(6,6),facecolor='lightblue')

colors = sns.color_palette('Set3')

ijm['JobType'].value_counts().plot.pie(autopct='%1.1f%%',shadow=True,explode=(0.05,0.05,0.05,0.05),
startangle=140, colors={'purple','violet','red','green'},wedgeprops={'edgecolor': 'black', 'linewidth': 1})

plt.title('Type of Job Distribution')

plt.ylabel('')

plt.show()
```

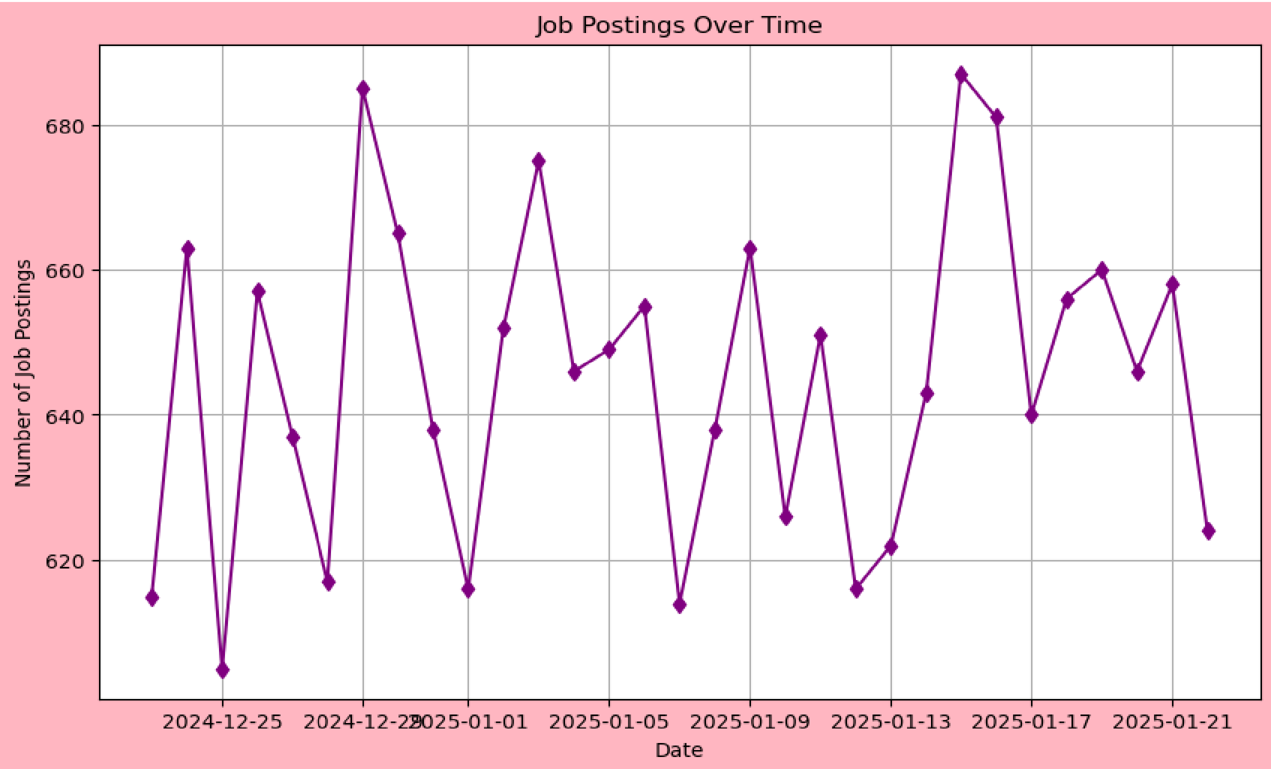
Output:



Plot Graphs:

```
ijm['Posted Date'] = pd.to_datetime(ijm['Posted Date'], dayfirst=True)

jobs_over_time = ijm.groupby('Posted Date').size()

plt.figure(figsize=(10,6),facecolor='lightpink')

plt.plot(jobs_over_time.index, jobs_over_time.values, marker='d', color='purple')

plt.title('Job Postings Over Time')

plt.xlabel('Date')

plt.ylabel('Number of Job Postings')

plt.grid(True)

plt.show()
```

Output:



Line Plot

#extracting avg salary

```
ijm['Salary Range'] = ijm['Salary Range'].str.replace(r'(\d+)\+', r'\1', regex=True)

ijm[['Min Salary', 'Max Salary']] = ijm['Salary Range'].str.replace('LPA', '').str.strip().str.split('-', expand=True)
```

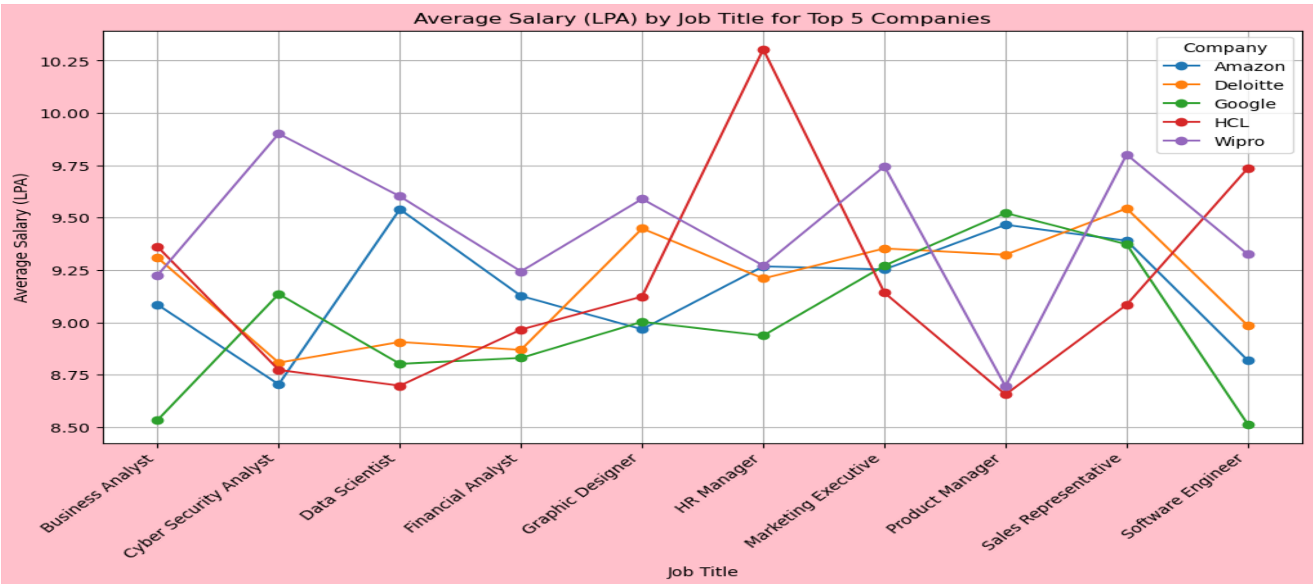
```
ijm['Min Salary'] = ijm['Min Salary'].astype(float)

ijm['Max Salary'] = ijm['Max Salary'].astype(float)

ijm['Average Salary (LPA)'] = (ijm['Min Salary'] + ijm['Max Salary']) / 2

grouped = ijm.groupby(['Company Name', 'Job Title'])['Average Salary (LPA)'].mean().reset_index()
```

#filtering top 5 companies

```
top_5_companies = ijm['Company Name'].value_counts().nlargest(5).index

filtered_df = ijm[ijm['Company Name'].isin(top_5_companies)]

#groupby comapny, job title and avg salary

grouped = filtered_df.groupby(['Company Name', 'Job Title'])['Average Salary (LPA)'].mean().reset_index()
```

```
plt.figure(figsize=(12,6),facecolor='pink')

for company in grouped['Company Name'].unique():

    company_data = grouped[grouped['Company Name'] == company]

    plt.plot(company_data['Job Title'], company_data['Average Salary (LPA)'], marker='o', label=company)

plt.title('Average Salary (LPA) by Job Title for Top 5 Companies')

plt.xlabel('Job Title')

plt.ylabel('Average Salary (LPA)')

plt.xticks(rotation=45, ha='right')

plt.legend(title='Company')

plt.grid(True)

plt.show()
```

Output:



Advantages:

- **Data-Driven Policy Making**:
  Enables governments to craft targeted employment schemes, labor reforms, and social security programs based on real workforce needs.

- **Sectoral Growth Insights**:
  Identifies booming industries like IT, green energy, and healthcare, helping direct investments and training efforts effectively.

- **Skill Gap Identification**:

  Reveals mismatches between available skills and industry demands, guiding curriculum updates and vocational training.

- **Regional and Demographic Mapping**:

  Highlights employment disparities across states, genders, and socio-economic groups, supporting inclusive development.

- **Forecasting Future Trends**:

  Anticipates shifts due to automation, AI, and remote work, allowing proactive workforce planning.

- **Business Intelligence**:

  Helps companies optimize hiring strategies, benchmark salaries, and understand labor availability across regions.

Conclusion:

In conclusion, the India Job Market analysis using Python, Matplotlib, and Seaborn presents a powerful and comprehensive approach for extracting valuable insights from skill development and building the carrer in the world of economy.

The integration of these versatile tools offers numerous advantages that contribute to data-driven decision-making and improved business performance for many organizations as well the unemployment reduction:

Organizational Strategy and Workforce Planning

- Job market data helps companies understand hiring trends, skill availability, and salary benchmarks.
- It enables smarter recruitment, talent retention, and workforce distribution across regions and sectors.

Business Growth and Competitiveness

- Organizations can align their operations with high-growth sectors like IT, green energy, and digital services.
- Data insights support innovation, cost optimization, and expansion into emerging markets.

Skill Development and Talent Management

- Analysis reveals current and future skill demands, guiding corporate training and upskilling programs.
- It ensures that employees remain relevant in a rapidly evolving job landscape.

Policy and Government Program Alignment

- Businesses can better leverage government initiatives like Skill India, PLI, and Make in India.
- Data helps evaluate the impact of these programs and align corporate goals with national priorities.

Economic Growth and Employment Elasticity

- A 1% increase in GDP has led to a 1.11% rise in employment between 2017–23, showing strong job responsiveness to economic growth.
- Employment growth boosts consumption, productivity, and overall economic resilience.

Inclusive and Sustainable Development

- Data highlights regional and demographic disparities, encouraging inclusive hiring and CSR initiatives.
- It supports long-term sustainability by promoting equitable access to opportunities.

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